TEXORA · LECTURE SHEET

3D Models of Fiber Reinforced Composites

Building the representative volume element: fibre packing geometry and volume fraction, the CAD and boolean workflow for fibre-in-matrix models, woven-composite cells, periodic boundary conditions, and checking FEA output against micromechanics.

ME-1202 · ENGINEERING DRAWING-II · TOPIC 4 · LEVEL 1 TERM 2

1. The representative volume element (RVE)

Nobody meshes a whole composite beam fibre by fibre — a square millimetre of carbon laminate contains thousands of 7 μm fibres. Instead we model one repeating block, the **representative volume element** (or unit cell): a prism of matrix with its fibres placed in the packing pattern, small enough to mesh finely, large enough to stand for the whole material. Simulate the RVE and its average response becomes the material property handed to the structural model — the same unit-cell thinking as the woven fabric of Topic 3, one scale down.

2. Packing geometry and fibre volume fraction

For parallel cylindrical fibres of diameter d on a square grid of pitch p (centre-to-centre spacing), one fibre occupies each p × p square, so

$$V_f = (\pi/4)(d/p)^2 \quad \text{(square array)}; \quad V_f = (\pi/(2\sqrt{3}))(d/p)^2 \quad \text{(hexagonal array)}$$

WORKED EXAMPLES

Carbon fibres, d = 7 μm, p = 10 μm: square $V_f = 0.785 \times (0.7)^2 =$ **38.5%**; hexagonal $V_f = 0.907 \times (0.7)^2 =$ **44.4%** — hexagonal packing always fits more fibre, and fibres touching (p = d) cap the two arrays at 78.5% and 90.7%. Real laminates live at 50–65%.

Counting fibres: a 10 × 1 mm laminate section at $V_f = 0.5$ holds 5 mm² of fibre; one 10 μm fibre has area $(\pi/4)(0.01\ \text{mm})^2 = 7.85 \times 10^{-5}$ mm²; $N \approx 5/(7.85 \times 10^{-5}) \approx$ **64,000 fibres** — the honest reason RVEs exist.

3. The modelling workflow

1. **Fibre bodies:** cylinders (or lenticular yarn bundles for woven models, Topic 3) created as separate solids, patterned by array to the packing plan.
2. **Matrix block:** a solid prism of the cell dimensions; use a **boolean cut** — subtract the fibres from the block — to produce the matrix with fibre-shaped tunnels, guaranteeing a perfectly mating interface.
3. **Reassemble:** keep the fibre solids and the cut matrix as separate bodies in one part/assembly; assign fibre properties (carbon: E ≈ 230 GPa along the axis — and note carbon is anisotropic itself) and matrix properties (epoxy: E ≈ 3 GPa).
4. **Mesh:** refine at the interface, where stress gradients live; a conformal (shared-node) mesh across the interface models a perfect bond; cohesive/contact definitions model debonding when that is the study.
5. **Boundary conditions: periodic boundary conditions** tie opposite faces of the cell so the block behaves as one tile of an infinite material — the standard correction to simply clamping faces, which over-stiffens the cell.

Woven composites raise the game: the "fibres" are yarn bundles following the interlaced paths of Topic 3, embedded in a matrix block; tools like TexGen generate them from weave data and export ready RVEs. **Short-fibre** models distribute fibre segments with statistical orientation — the model now has an orientation distribution function, not a single direction, and its properties come out nearly isotropic when the distribution is random.

4. Checking the simulation against micromechanics

The RVE must first reproduce the textbook limits before it is trusted anywhere else: loaded along the fibres its modulus must match the rule of mixtures $E_c = E_fV_f + E_mV_m$ (ME-1207, Topic 4); across them it should approach the inverse-rule value. Extract the RVE's average stress and strain from the FEA (reaction force ÷ face area; applied displacement ÷ cell length) and compare: agreement within a few percent validates mesh and boundary conditions. Only then do the interesting studies begin — the stress concentration in the matrix between close fibres, the first debond site, thermal residual stresses from curing cool-down.

5. Key points

- RVE = one repeating fibre-packing block whose averaged response is the material's property.
- V_f from geometry: square $(\pi/4)(d/p)^2$, hexagonal $(\pi/(2\sqrt{3}))(d/p)^2$; ceilings 78.5% / 90.7%; practice 50–65%.
- Workflow: pattern fibre solids → boolean-cut the matrix → separate bodies, real materials → interface-refined mesh → periodic BCs.

- Woven-composite RVEs use yarn paths in matrix; short-fibre RVEs use orientation statistics.
- Validate every RVE against the rule of mixtures before believing its local stresses.

6. Practice

Answer in your own words; brief model answers follow.

1. Glass fibres of diameter 14 μm are packed hexagonally at a pitch of 18 μm. Find V_f.
2. A laminate section of 5 mm² total area must reach $V_f = 0.6$ with 7 μm fibres. Estimate the number of fibres in the section.
3. Explain why periodic boundary conditions give a truer RVE stiffness than rigidly fixing one face and pulling the other.
4. Your RVE returns a longitudinal modulus 15% below the rule-of-mixtures value. List three modelling faults to check before blaming the formula.
5. How does the RVE of a woven composite differ from that of a unidirectional one, and what extra failure site appears in it?
6. Why must the boolean cut for the matrix be made from the very fibre solids used in the assembly, rather than from a separately drawn set of the same nominal size?

Answer key

1. $V_f = (\pi/(2\sqrt{3}))(d/p)^2 = 0.907 \times (14/18)^2 = 0.907 \times 0.605 = 0.549 \approx 55\%$ — a realistic laminate value.
2. Fibre area total = $0.6 \times 5 = 3$ mm². One fibre: $(\pi/4)(0.007\text{ mm})^2 = 3.85 \times 10^{-5}$ mm². $N = 3/(3.85 \times 10^{-5}) \approx 78{,}000$ fibres.
3. Fixing faces forces every point of the face to move as a rigid plate, suppressing the local waviness with which a real neighbourhood of material would comply — the cell answers too stiff and its edge stresses are fictitious. Periodic conditions instead force opposite faces to deform as identical copies of one another, exactly as neighbouring identical cells constrain each other in the real infinite material; the interior stress state is then honest.
4. (1) The fibre volume fraction in the model is not what you think — measure the actual solid volumes; a pitch error of 8% costs about 15% of V_f. (2) The load is not purely axial — misaligned faces or a bending component waste stiffness. (3) The interface mesh is not conformal/bonded — gaps or sliding contact soften the transfer of load into the fibres. (Also check material cards: entering the transverse modulus of carbon, ≈ 15 GPa, as the axial one is a classic.)
5. The UD RVE is a straight prism: parallel fibres in matrix, one symmetry-rich cell. The woven RVE's reinforcement follows interlaced, undulating yarn paths (lenticular bundles over and under each other) inside a larger cell spanning the weave repeat. The extra failure site is the yarn cross-over:

resin-rich pockets and sharp yarn curvature there concentrate stress and start matrix cracks and delamination.

6. Because the interface must coincide exactly. A separately drawn “identical” set inevitably differs by rounding, tolerance stacking or a stale dimension, leaving micron-scale gaps or overlaps at the interface; the mesh then fails, or worse, succeeds with a fictitious debond or interpenetration built in. Cutting with the actual solids guarantees one shared surface — the definition of a bonded interface.
